Data Sources & Reproducibility

Shared appendix for the electoral-health series — `who-decides-washington.md`, `safe-seat-washington.md`, `cross-state-fec-money.md`, and the parent `electoral-health-whitepaper.md`. Every input is a public record; this document lists each source and how to obtain it, so the studies are reproducible from primary sources rather than from a redistributed dataset.

Reproducibility statement

These studies are **source-reproducible**: an independent researcher can obtain every input directly from the issuing agency (links below) and re-run the analysis using the public code repository, and should arrive at materially the same figures. They are **not bitwise-reproducible**, for two ordinary reasons that any voter-file study shares:

1. **The voter file is a living snapshot.** State voter rolls and cumulative vote-history files change continuously (registrations, cancellations, address changes, each new election appended). A request filed on a later date returns a slightly different roll, so exact counts will differ at the margin. Cite the **access date** (below) when reporting.
2. **Processing must be re-run, not copied.** Name/county normalization, the precinct-namespace crosswalks, and the voter↔donor matcher (`match_voters_to_donors`, with its uniqueness-guarded tiers) must be executed on the freshly obtained data. All of this logic is in the public repository.

We therefore publish **citations + code, not data** — which is also the only lawful option for the voter files (see *Use restrictions*).

Code: all loaders, the matcher, and the `scripts/diag_*.py` analysis scripts that produce every figure in these papers are in the public reproduction repository (`github.com/skirby359/who-decides`). Each paper names the exact script under its figures (e.g. `scripts/diag_safe_seat_states.py`, `scripts/diag_wa_individual_findings.py`, `scripts/diag_cross_state_money_matrix.py`).

Sources by paper

Washington — voter file, vote history, elections, money

Asset	Issuing agency	Access	Used in
Statewide Voter Registration Database (VRDB) — registrants + cumulative vote history (~5.5M voters, ~27.1M vote records, ~100% **year of birth** — not full date of birth; stored as a July-1 sentinel)	WA Secretary of State	Standard statewide **VRDB extract** (the single public extract; regenerated monthly) via the SoS extract-request form (`sos.wa.gov/washington-voter-registration-database-extract`). Includes year of birth by statute — full DOB withheld (RCW 29A.08.710); use restricted to political/scholarly purposes, not redistributable (RCW 29A.08.720; penalties .740). *April 2026 extract, requested April 8 2026; vote history through the Feb 2026 special.*	Who Decides WA; cross-state §F

Asset	Issuing agency	Access	Used in
Certified statewide summary returns, 2014–2025 (`*_AllState.csv`) — one row per candidate per race, all seats. **This, not the precinct file, is the safe-seat universe.** The precinct exports (`*_AllStatePrecincts.csv`) are loaded for other papers but are NOT the seat universe: King County is absent from them for 2016 and 2018, which cost 24 House seats a cycle and is the defect that forced this paper's rebuild.	WA Secretary of State	`results.vote.wa.gov` per-election exports (e.g. `…/results/20241105/export/20241105_AllState.csv`)	Safe-Seat WA
Frozen WA party-string mapping — all 32 distinct "Prefers … Party" strings across the five certified generals, classified explicitly under both the published and the loose specification, each with a reason	derived from the certified files above; author-adjudicated 2026-08-08	`docs/reference/wa_party_strings_2016-2024.csv`	Safe-Seat WA
Campaign-finance contributions	WA Public Disclosure Commission	PDC open data via `data.wa.gov` (Socrata); statewide contributions + the Campaign Finance Summary dataset (`3h9x-7bvm` , full filer universe: candidates, party/caucus committees, PACs, ballot committees). Portal: `pdc.wa.gov/political-disclosure-reporting-data`	cross-state money (WA PDC layer)
King County precinct crosswalk (`cidw-fyff`)	King County Elections / Open Data	`data.kingcounty.gov`	VRDB precinct bridging

Federal money — all states (FEC)

Asset	Source	Access	Used in
Individual contributions (`indiv{yy}.txt`), per cycle 2018–2026	Federal Election Commission	`fec.gov/files/bulk-downloads/`	cross-state §A–I; inflow
Committee master (`cm{yy}.zip`) + **Candidate master** (`cn{yy}.zip`)	FEC	same bulk portal	committee→candidate→office-state resolution (§G); inflow recipient anchoring
FEC REST API (supplemental per-candidate pulls)	FEC	`api.open.fec.gov/v1` (free API key)	finance backfills

State-level campaign finance is loaded for all four states (WA PDC, NY State Board of Elections via `etl/adapters/ny_finance.py` , ID Sunshine, TX Ethics Commission via `etl/adapters/tx_tec.py`) into the shared `candidate_finance` table, but the cross-state money cuts in these papers are federal; the state layer is available for follow-on state-level analysis.

Recipient-anchored inflow (`fec_inflow.duckdb`) is the FEC bulk `indiv` files filtered to committees whose connected

candidate's office state is WA/NY/TX/ID — built by `scripts/load_fec_inflow_bulk.py`. Idaho FEC contributions (outflow, donor `state='ID'`) and inflow were added 2026-07-19, bringing ID to full federal parity with the other three.

New York

Asset	Source	Access	Used in
Precinct/county election results (incl. Assembly & Senate, 2022)	NYS Board of Elections, via **OpenElections-NY**	`github.com/openelections/openelections-data-ny`	Safe-Seat four-state map
Assembly District 23, 2022 general — certified contest	NYS Board of Elections (direct)	`results.elections.ny.gov/contest/116`, retrieved 2026-08-08 → pinned at `docs/reference/ny_ad23_2022.csv`	Safe-Seat four-state map
NY federal contributions	FEC bulk (donor `state='NY'`)	as above	cross-state money
NYSVOTER statewide voter file — 13.54M registrants with **individual party enrollment**, **full date of birth**, status code, and vote history *(loaded via `scripts/load_ny_voters.py` → `data/ny_vrdb.duckdb`)*	NYS Board of Elections	Public-records request under FOIL (N.Y. Pub. Off. Law art. 6), subject to the Board's elections-purpose certification. *Production `ALLNYVOTERS20260629.zip`, dated 2026-06-29.* The delivered columns are a property of this production, not a documented statutory layout — the Board's public request page does not publish the file layout. 91.9% of records are `status_code='A'`. Not redistributed.	Who Decides NY; donor-class F1/F3/F4
NYSBOE contribution disclosure feed — transaction-level, 12.6M rows back to 1999; 3,954,090 individual Schedule A rows / $880.3M loaded for cycles 2018–2026 *(via `scripts/load_ny_contributions.py`)*	NYS Board of Elections, via data.ny.gov	Socrata dataset **`4j2b-6a2j`**. Itemization threshold: contributor aggregate **> $99** (N.Y. Elec. Law § 14-102); $99 or less reportable in the aggregate.	donor-class state panel

Texas

Asset	Source	Access	Used in
VTD-level general-election returns, 2016–2024	**Texas Legislative Council** Capitol Data Portal	`data.capitol.texas.gov` (CKAN; exact resource URLs in `scripts/download_tx_tlc.py`)	Safe-Seat
r206 "Election24G" by-district reports (President/US-Sen per House/Senate/Congressional district; cover *every* district incl. uncontested)	Texas Legislative Council	same portal (`planh2316_…`, `plans2168_…`, `planc2333_r206_election24g.xls`)	TX safe-seat backfill (§ four-state map)

Asset	Source	Access	Used in
TX federal contributions	FEC bulk (donor `state='TX'`)	as above	cross-state money

Why the TX backfill is needed and sound: TLC's canvass-grade VTD returns omit uncontested races (no precinct tally is published when a seat is unopposed), so 96/150 House districts appear in the VTD file. The 54 absent seats are verified seat by seat against the Texas Secretary of State's certified results, which independently show exactly those 54 districts as single-candidate and supply each one's observed winning party.

That method replaced an earlier one, and the replacement changed a published figure — which is why the old description is recorded here rather than simply overwritten. The earlier backfill cross-checked the 54 against a press-reported unopposed list and used the r206 report's presidential lean to label each seat's holding party. Lean is a proxy for the winner, not the winner, and it was wrong often enough to matter: the certified split is 56 D / 85 R, against 51 D / 90 R under the lean proxy. Do not reintroduce the press-list-plus-lean method; Appendix F of the safe-seat paper documents the correction.

Idaho

Asset	Source	Access
Statewide & county election results — **general + primary canvasses, 2016–2024** (loaded via `scripts/download_id_sos.py`; primaries added 2026-07-09)	Idaho Secretary of State	`sos.idaho.gov/elections/data/results/`; archive `archive.sos.idaho.gov/ELECT/results/`; some county PDFs (e.g. Bannock County)
State campaign-finance contributions	Idaho SoS **Sunshine** portal	`sunshine.voteidaho.gov` (export API `api-sunshine.voteidaho.gov`)
Federal contributions (donor `state='ID'`) + ID inflow	FEC bulk	as in the Federal-money section (added 2026-07-19)
Voter file w/ party affiliation + age + per-election ballot record *(loaded; powers "Who Decides Idaho?")*	Idaho SoS Elections	`forms.sos.idaho.gov/voter-registration-request-form/` ($20/report). Public statewide list per **Idaho Code § 34-437A(3)** — carries name, street + mailing address, county, gender, **age (not DOB)**, declared party, and participation history; DOB and other registration-card data are protected separately. (**Idaho Code § 74-120** is the unrelated mailing/telephone-list restriction.) *Extract received 2026-07-01.* The export carries **no active/inactive flag** — it is a current-roll extract and `load_id_voters.py` sets every row `status_code='A'`, so active-only and all-record baselines coincide by construction.

Idaho primary coverage: 2024/2022/2018/2016 include the legislative + congressional primaries (each party's contest is a distinct race); the 2020 primary is statewide-only (US Senate + US House) — that COVID-era mail-only cycle's legislative results were published by the SoS at county level only, with no precinct-level file.

Auxiliary (warehouse-wide; not core to these three papers)

Census ACS 5-yr (`api.census.gov`; ACS 2020–2024 table B01001 → WA adult-resident age composition in Who-Decides-WA, via `scripts/acs_wa_adult_age.py`), Census geocoder + TIGER shapefiles (`www2.census.gov/geo/tiger`), BLS unemployment (`api.bls.gov`), WA precinct shapefiles (`sos.wa.gov`).

Use restrictions (why we cite, not redistribute)

- **WA VRDB** — voter-registration data is obtainable but its use is restricted to non-commercial / political/scholarly

purposes (WA **RCW 29A.08.720** — the lists "may be used for any political purpose" but not for advertising or solicitation; criminal/civil penalties at **RCW 29A.08.740**). Not redistributable. The standard public extract carries **year of birth, not full date of birth**: WA withholds full DOB as protected information (**RCW 29A.08.710**), and under **2026's SB 5892** an election official who discloses a full birthdate faces a class C felony. So no full DOB was ever obtainable via this extract.

- **ID voter file** — political/scholarly use only; no use as a mailing or telephone list without consent (**Idaho Code §74-120**, which releases registrant age but withholds date of birth, driver's license number, and — on a showing of good cause — residence address).
- **NY voter file** — FOIL access for elections purposes; aggregate outputs only.
- **FEC, TLC, SoS election results, PDC, Sunshine** — fully public, redistributable.

Because the voter files carry PII and cannot be redistributed, the studies report only aggregates and publish the access path, not the data. (The product layer additionally enforces a PII firewall — `src/wa_analyzer/product/firewall.py`.) *Statute citations verified against the current code 2026-06-29 (WA RCW 29A.08.720/.740; Idaho Code §74-120); confirm once more at formal-publication time.*

How to replicate

1. Obtain the inputs above (file the records requests; download the public bulk files).
2. Clone the repository and load each state per its loader (`README.md` / `CLAUDE.md`).
3. Build the recipient-anchored inflow: `scripts/load_fec_inflow_bulk.py` .
4. Run the matcher. In the public release it ships standalone as `scripts/donor_matcher.py` (duckdb + stdlib only), driven by the per-state wrappers: `python scripts/match_id_voters_to_donors.py --source fec --tiers full` . `--tiers full` is the primary specification — the full-first-name key alone — and is the default; `--tiers all` reproduces the superseded all-tier specification. Restricting the match is NOT equivalent to filtering a built panel on `match_quality` (identical donor counts, dollar totals 3.8–9.4% apart), so pick one and stay in it.
5. Re-run the per-paper scripts named under each figure.

Report your VRDB/voter-file **access dates** alongside any replicated counts.